DEEP RESEARCH Prompt:

Scientist Information

- **Name:** Mitchell Guttman
- **Dates:** 1982-
- **Era/Field:** Epigenetics & Genomics
- **Course context:** BIOL 3030 intro biochemistry audience; collaborative class eBook; final narrative will be ~500 words written by the student

---

Output Format (Strict)

A) ANCHOR SCENES (3–6)

- **SCENE #1: First Day in Guttman Lab**
  - Summary: Mitchell Guttman begins a new horizon as the principal investigator of his own lab at Caltech, but he begins to struggle with the 'intellectual loneliness' of the new setting.
  - Date/Place: Guttman Lab, California Institute of Technology, June 2013
  - People: Mitchell Guttman
  - Concrete details:"the thing that really caught me off guard, was how intellectually lonely it is to start a lab…All of a sudden, the basic things become harder and that's not even to mention starting from scratch and having to teach everyone who comes into the lab"
  - Why it matters: Guttman described this as his greatest challenge when starting his own lab,
  - Evidence quality: primary - from Guttman himself
  - Myth risk: This underestimates his desire to be a principal investigator, highlighting a negative aspect of the lab without completely describing the good.
  - Sources: [Cell scientist to watch – Mitchell Guttman | Journal of Cell Science | The Company of Biologists]
- **SCENE #2: Lander helps Guttman land on lncRNA**
  - Summary: Eric Lander mentored Guttman in Lander's lab, providing him with the equipment and advice to study lncRNA that would shape Guttman's career as a researcher.
  - Date/Place: Broad Institute at MIT, circa 2007-2011
  - People: Mitchell Guttman, Eric Lander
  - Concrete details: Eric Lander was the director of Broad Institute at MIT and Guttman's graduate advisor. Guttman worked in his lab, where Lander told him "the famous Wayne Gretzky quote: go after problems that everyone will be interested in, rather than problems that everyone is currently interested in." This helped Guttman decide lncRNA would become the scope of his research. Lander's lab also had rare laboratory equipment that could analyze histone modifications.
  - Why it matters: Without Lander's help, Guttman likely would not have decided on lncRNA for his focus of study. Guttman's lncRNA research is the focal point of his success, winning him several publications and awards.
  - Evidence quality: primary - from Guttman himself
  - Myth risk: This perspective provides a lot of credit to Lander and makes him seem too perfect.

    - Sources: Minding the Gaps in the Genome: An Interview with Mitch Guttman - This is Caltech Cell scientist to watch – Mitchell Guttman | Journal of Cell Science | The Company of Biologists
- **SCENE #3: lncRNA Discovery**
    - Summary: Guttman makes an important discovery regarding lncRNA that opened the doors for possible study as a researcher.
    - Date/Place: Lander Lab at Broad Institute at MIT, circa 2007
    - People: Mitchell Guttman, Eric Lander, other lab members
    - Concrete details: Lander Lab accumulated lots of unpublished data around histone modification. That is when Lander told Guttman,"I bet there's something here to be found. You're a computational guy; why don't you play around with it?" "The first thing I did, as a good computational guy, was to try to figure out a good algorithm to make sense of it. Once I did, it kind of hit me in the face," says Guttman. He observed lots of lncRNA, in particular, how they had the same protein-coding patterns as genes but did not code for proteins. From there, Guttman realized the opportunity to study how lncRNA governs biological processes.
    - Why it matters: This observation was the pivotal moment that inspired Guttman to pursue studying lncRNA as his study of research that is responsible for his prestige as a researcher and biochemist.
    - Evidence quality:primary - from Guttman himself
    - Myth risk: This seems too good to be true, however, there are plenty of discoveries that happened seemingly by accident, so it is not too crazy.
    - Sources: Minding the Gaps in the Genome: An Interview with Mitch Guttman - This is Caltech Mitchell Guttman, Ph.D. - The Sontag Foundation
- **SCENE #4: Winning NIH Director's Early Independence Award**
    - Summary: Mitchell Guttman wins the NIH Director's Early Independence Award
    - Date/Place: Caltech, 2012
    - People: Mitchell Guttman
    - Concrete details: This award is granted to "promising, newly graduated scientists with the intellect, scientific creativity, drive, and maturity bypass the traditional postdoctoral training period to launch independent research careers." This is granted to a non-independent researcher for $250,000 per year for 5 years.
    - Why it matters: Guttman is awarded a significant financial compensation as well as recognition for his work as a non-independent researcher, which grants him the opportunity to become an independent researcher and further advance his studies, likely pivotal to him starting his own lab.
    - Evidence quality:Information regarding the award is directly from the NIH, however, there is
    - Myth risk: Though this is a significant award, it is not explicitly stated whether this truly incited a turning point in his career.
    - Sources: NIH Director's Early Independence Award Program - 2012 Award Recipients | NIH Common Fund NIH Director's Early Independence Award | NIH Common Fund

B) DIRECT QUOTES (3–7)

1. "these large noncoding RNAs that looked like proteins but didn't act like proteins—they did something else. What that something else was, I didn't know, but it was new and unexplored and was clearly important. The potential was huge.."
    - Context/date: from transcript of Caltech interviewing Mitchell Guttman, October 8, 2013

    - Source (full citation): Caltech. (2013, October 8). *Minding the gaps in the genome: An interview with Mitch Guttman*. https://www.caltech.edu/about/news/minding-gaps-genome-interview-mitch-guttman-40705
2. "What exactly I wanted to do, how exactly I would do it, what it would entail was not at all obvious to me, and in hindsight I had no idea."
    - Context/date: Guttman's guest appearance on Epigenetics Podcast February 22, 2024
    - Source (full citation): Active Motif. (2024, February 22). Split-Pool Recognition of Interactions by Tag Extension (SPRITE) – Mitch Guttman (Ep. 119) [Audio podcast episode]. Epigenetics Podcast. https://www.activemotif.com/podcasts-mitch-guttman
3. "My first interest in biology started when I was in elementary school…when Dolly the sheep was cloned… it was the first time I really understood the power, opportunities, the transformative potential of biology."
    - Context/date: Guttman's guest appearance on Epigenetics Podcast February 22, 2024
    - Source (full citation): Active Motif. (2024, February 22). Split-Pool Recognition of Interactions by Tag Extension (SPRITE) – Mitch Guttman (Ep. 119) [Audio podcast episode]. Epigenetics Podcast. https://www.activemotif.com/podcasts-mitch-guttman
4. "For me, collaborative science is what makes science fun; being able to leverage different talents of people and expertise that are complementary to your own to do something even better than you could do on your own."
    - Context/date: Archive of an interview with Guttman in the Journal of Science, May 2017
    - Source (full citation): The Company of Biologists. (2017). Cell scientist to watch – Mitchell Guttman. Journal of Cell Science, 130(9), 1497–1499. https://doi.org/10.1242/jcs.203976
5. "The Sontag Foundation provided my lab with critical funding to explore the fundamental principles of how lncRNAs work in normal cells, so we can understand why their disruption leads to various human cancers and utilize this knowledge to develop specific therapeutics in cancer treatment."
    - Context/date: Award speech for 2014 Sontag Award in Sontag Foundation archive
    - Source (full citation): Sontag Foundation. (n.d.). Mitchell Guttman, Ph.D. https://sontagfoundation.org/grantee/mitchell-guttman-phd/

C) HUMAN REALITY DETAILS (8–15 bullets)

- Computational biologist — Source: https://thisis.caltech.edu/news/minding-gaps-genome-interview-mitch-guttman-40705
- Always interested in science — Source: https://thisis.caltech.edu/news/minding-gaps-genome-interview-mitch-guttman-40705
- Enjoys collaboration — Source: https://journals.biologists.com/jcs/article/130/9/1497/56750/Cell-scientist-to-watch-Mitchell-Guttman
- reflective — Source: https://journals.biologists.com/jcs/article/130/9/1497/56750/Cell-scientist-to-watch-Mitchell-Guttman
- While still a graduate student at the Broad Institute, Guttman led the team that first described a special class of genes called lncRNAs — Source: Minding the Gaps in the Genome: An Interview with Mitch Guttman - This is Caltech
- when I was a sophomore [age 15 or 16] in high school, my chemistry teacher encouraged me to take it a step further and explore scientific research. — Source: Cell scientist to watch – Mitchell Guttman | Journal of Cell Science | The Company of Biologists
- I had the opportunity to work with a clinical pathologist and a basic cancer researcher— Source: Cell scientist to watch – Mitchell Guttman | Journal of Cell Science | The Company of Biologists
- Had difficulty balancing work and life — Source: Cell scientist to watch – Mitchell Guttman | Journal of Cell Science | The Company of Biologists

- Likes to keep a positive attitude — Source: Cell scientist to watch – Mitchell Guttman | Journal of Cell Science | The Company of Biologists
- Excited by his discoveries — Source: https://journals.biologists.com/jcs/article/130/9/1497/56750/Cell-scientist-to-watch-Mitchell-Guttman

D) SCIENCE CONNECTION (Intro Biochem)

- Best-fit topic: Genomics
- Precise contribution (1 sentence): Guttman continues to discover the biological role of long non-coding RNA (lncRNA) as structural components of chromosomes and regulators of gene expression, particularly in the context of driving inhibitive Xist properties and tumor suppressing pathways in cancer.
- Bridge explanation (2–4 factual sentences): lncRNAs are long RNA transcripts with a modular structure with repetitive sequences that are highly cell type and state specific. They are able to form condensates that organize regulatory molecules as well as structurally support chromatin loops. Additionally, lncRNA interacts with significant chromatin-modifying complexes in order to activate or inactivate gene expressions.
- Scientist's own framing/analogy (if found):
- Sources: Long non-coding RNAs: definitions, functions, challenges and recommendations | Nature Reviews Molecular Cell Biology

E) COMPLICATIONS OR CONTROVERSIES (If any)

- Item: Guttman 'discovered lncRNA
  - What's agreed: he discovered the significant biological roles it can play
  - What's disputed: He was not the first person to discover their existence
  - Sources:Jarroux, J., Morillon, A., & Pinskaya, M. (2017). History, Discovery, and Classification of lncRNAs. Advances in experimental medicine and biology, 1008, 1–46. https://doi.org/10.1007/978-981-10-5203-3_1

    Minding the Gaps in the Genome: An Interview with Mitch Guttman - This is Caltech

F) "NOT ON WIKIPEDIA" FINDS (5+)

- Worked under a clinical pathologist as a high school student— It helped him develop a deep understanding of the research field at a relatively young age — Source: Cell scientist to watch – Mitchell Guttman | Journal of Cell Science | The Company of Biologists
- His family has no scientific background— He found a love for biology on his own and charted his own path — Source: Cell scientist to watch – Mitchell Guttman | Journal of Cell Science | The Company of Biologists
- He wrote a computer program called Scripture to analyze non-coding protein regions — He incorporated his computational skills to propel his research, prioritizing his individual skillset — Source: Minding the Gaps in the Genome: An Interview with Mitch Guttman - This is Caltech
- developed SPIDR (split-and-pool identification of RBP targets), a highly multiplexed split-pool method that profiles the binding sites of dozens of RBPs simultaneously— He also develops new approaches to lncRNA analysis — Source: Wolin, E., Guo, J. K., Blanco, M. R., Perez, A. A., Goronzy, I. N., Abdou, A. A., Gorhe, D., Dong, W., Keskin, A., Valenzuela, E., Abdou, A. A., Urbinati, C. R., Rube, H. T., Querido, J. B., Guttman, M., & Jovanovic, M. (2025). SPIDR enables multiplexed mapping of RNA–protein interactions and uncovers a mechanism for selective translational suppression upon cell stress. Cell, 188(19), 5384–5402.e25. https://www.cell.com/cell/article/S0092-8674(25)00743-3/fulltext

- Developed ChIP-DIP that maps the positions of several DNA proteins — It shows him applying his computational background to create new approaches to analyzing DNA and the genome — Source: https://www.caltech.edu/about/news/new-technique-maps-hundreds-of-proteins-simultaneously-within-cell-nuclei

G) SOURCES CONSULTED (Full citations)

1. Caltech. (2024, December 18). New technique maps hundreds of proteins simultaneously within cell nuclei. https://www.caltech.edu/about/news/new-technique-maps-hundreds-of-proteins-simultaneously-within-cell-nuclei
2. Wolin, E., Guo, J. K., Blanco, M. R., Perez, A. A., Goronzy, I. N., Abdou, A. A., Gorhe, D., Dong, W., Keskin, A., Valenzuela, E., Abdou, A. A., Urbinati, C. R., Rube, H. T., Querido, J. B., Guttman, M., & Jovanovic, M. (2025). SPIDR enables multiplexed mapping of RNA–protein interactions and uncovers a mechanism for selective translational suppression upon cell stress. Cell, 188(19), 5384–5402.e25. https://www.cell.com/cell/article/S0092-8674(25)00743-3/fulltext
3. Mitch Guttman (Ep. 119) [Audio podcast episode]. Epigenetics Podcast. https://www.activemotif.com/podcasts-mitch-guttman
4. Sontag Foundation. (n.d.). Mitchell Guttman, Ph.D. https://sontagfoundation.org/grantee/mitchell-guttman-phd/
5. Caltech. (2013, October 8). Minding the gaps in the genome: An interview with Mitch Guttman. https://www.caltech.edu/about/news/minding-gaps-genome-interview-mitch-guttman-40705
6. The Company of Biologists. (2017). Cell scientist to watch – Mitchell Guttman. Journal of Cell Science, 130(9), 1497–1499. https://doi.org/10.1242/jcs.203976
7. Mattick, J.S., Amaral, P.P., Carninci, P. et al. Long non-coding RNAs: definitions, functions, challenges and recommendations. Nat Rev Mol Cell Biol 24, 430–447 (2023). https://doi.org/10.1038/s41580-022-00566-8
8. National Institutes of Health. (2012). NIH Director's Early Independence Award: 2012 award recipients [Archived web page]. In NIH Common Fund. Wayback Machine. https://web.archive.org/web/20200213142134/https://commonfund.nih.gov/earlyindependence/AwardRecipients12
9. National Institutes of Health. (n.d.). NIH Director's Early Independence Award. NIH Common Fund. https://commonfund.nih.gov/earlyindependence

H) FLAGGED "MYTH-RISK" DETAILS (If any)

List any dramatic details that are frequently repeated but weakly sourced.

- Claim:
- Where it appears:
- Why sourcing is weak:
- Best available evidence (if any):

LENS RECOMMENDATION (Do this only after completing A–G)

Lens scoring rubric (0–3 each)

Score based on how well your collected evidence supports a scene-driven narrative.

Eureka (Discovery crystallized)

- 3: Documented discovery moment with concrete details and strong sourcing
- 2: Clear discovery story but details thin or second-hand
- 1: Only broad “discovered X” claims
- 0: No identifiable discovery moment

Failure (Setback that changed trajectory)

- 3: Specific, well-documented setback that clearly shapes what happens next
- 2: Struggle exists but not clearly pivotal
- 1: Generic “it was hard”
- 0: None

Mentor (Relationship that redirected them)

- 3: Specific mentorship interaction or relationship pivot with clear before/after
- 2: Mentorship exists but pivotal moment is fuzzy
- 1: Only “worked with…”
- 0: None

Controversy (Conflict over credit, ideas, ethics, institutions)

- 3: Clear dispute with parties, stakes, and outcome; preferably multiple perspectives
- 2: Tension exists but one-sided or unclear
- 1: Vague conflict claims
- 0: None

Human Detail (Revealing personal fact that changes how we read the science)

- 3: Well-sourced detail that illuminates character and connects back to their work
- 2: Interesting but not connected to the scientific story
- 1: Trivia
- 0: None

Last Experiment (Final act, handoff, or late-life scene with closure)

- 3: Strong late-career or final-scene moment with concrete details and good sourcing
- 2: Late career story exists but not scene-worthy
- 1: Only death/retirement facts
- 0: None

Decision rule

- Choose the lens with the highest score.
- If tied, choose the lens tied to the most vivid, best-sourced anchor scene.
- Provide one backup lens only if it is within 1 point of the top score.

Output for lens recommendation

- Score table (lens: score 0–3 + 1 sentence justification)

The lens recommendation is Mentor because Guttman highly credits his mentor, of whom there is a detailed recount of guiding Guttman to a significant discovery.

- Primary recommended lens:
    - Why it fits (cite which scene card numbers support it): There are many direct quotes in scenes 3 and 4 that evidence a moment of mentorship that shapes his future career.
    - Best opening scene (scene #): scene 3
    - One quote to use early (quote #): quote 1
    - Cautions (thin sourcing, disputed detail, myth risk): It seems too good to be true that he stumbled upon something so significant so easily.
- Backup lens (if applicable): Eureka
    - He had a very vivid moment where he discovered the potential of lncRNA.
    - Scene 4
    - Quote 1
    - This was not necessarily a moment of new discovery, but rather a moment in which he found potential for a new discovery. This part would be rather thinly sourced, as it is not overly clear what he used to make this discovery, as simply ‘making an algorithm’ is rather broad.

Final Note

Do not write the profile. Deliver only evidence, citations, and lens recommendation after evidence collection.